



Ropener Curtain Controller

User Guide

Firmware 2.4.0

Ropener is a Wi-Fi connected curtain/cover controller. It pulls a poly rope with a small stepper motor to open and close your curtains. You can control it from any web browser on your network — no Home Assistant or cloud account required — or pair it with Home Assistant if you use one. It supports a daily open/close schedule (by fixed clock times or sunrise/sunset), automatic position calibration, and several tuning options.

1. At a Glance

- **Control from a browser:** open the device's web page on your phone or computer.
- **Three ways to operate:** the web page, the physical buttons on the device, or Home Assistant.
- **Open / close / partial:** drive the curtain fully open, fully closed, or to any position in between.
- **Daily schedule:** open and close automatically every day — at fixed clock times, or following sunrise and sunset.
- **Self-calibrating:** the device finds the fully-closed position by itself ("homing").

2. First-Time Setup: Connect to Wi-Fi

The device ships without any saved Wi-Fi network, so the first thing to do is tell it which network to join. There are two ways to do this.

Option A — The setup hotspot (recommended)

- Power on the device and wait about a minute.
- On your phone or laptop, open the Wi-Fi network list. You will see a network named “ropener-XXXXXX”, where XXXXXX is a unique code for your device.
- Write down or take a photo of this full name now — you will need the same “ropener-XXXXXX” to open the device in your browser later (see Section 3). The hotspot disappears once setup is done, so note it before continuing.
- Connect to it. A setup page should open automatically (this is the “captive portal”). If it does not, open a browser and go to <http://192.168.4.1>.
- Choose your home Wi-Fi network from the list, enter its password, and save.
- The device will reboot and join your network. The setup hotspot disappears once it connects successfully.

Remember: the “ropener-XXXXXX” code is also your device's web address — you'll type <http://ropener-XXXXXX.local> into a browser to control it. Keep the code handy.

Option B — Over USB (Improv)

If the device is plugged into a computer over USB, you can provision it from a web tool that supports “Improv” (for example, web.esphome.io). Open the tool, connect to the device's serial port, and enter your Wi-Fi details when prompted.

Note: Your Wi-Fi details are stored on the device and kept across reboots and firmware updates. You only need to do this once (unless you reset Wi-Fi — see Section 9).

3. Open the Web Control Panel

Once the device is on your network, open a web browser on the same network and go to:

<http://ropener-XXXXXX.local>

Replace XXXXXX with your device's code (the same code shown in the setup hotspot name). The page lists every control and reading described in this guide. Bookmark it for easy access.

Tip: If the .local address does not load, your network may not support it. In that case, find the device's IP address in your router and use <http://<that-ip>> instead.

4. Everyday Use

From the web page or Home Assistant

- **Open:** raises the cover to the fully-open position.
- **Close:** lowers the cover to the fully-closed position.
- **Set a position:** drag the cover slider to any point from 0% (closed) to 100% (open).

- **Stop:** halts movement immediately and leaves the cover where it is.

Using the physical buttons

The device has three buttons. While the cover is moving, a press of Button 1 or Button 2 stops it immediately. You can also press and hold a button to move the cover only while you hold it — it stops the moment you let go, and stops on its own at the fully open or closed end. This keeps it safe around children and the elderly: the cover never keeps moving on its own.

Button	Action	What it does
Button 1	Tap	Closes the cover (or stops it if moving).
Button 1	Press and hold	Closes only while held; stops on release and at the closed end.
Button 1	Seven quick taps	Starts homing / calibration (see Section 5).
Button 2	Tap	Opens the cover (or stops it if moving).
Button 2	Press and hold	Opens only while held; stops on release and at the open end.
Button 1 + Button 2	Hold both 3 sec	Sets the current position as home / 0% (see Section 5).
Button 3	Long press (3–10 sec)	Clears saved Wi-Fi and reboots into setup (see Section 9).

5. Homing (Calibrating the Closed Position)

Important — you must stop homing manually (automatic stop is disabled)

The sensor that normally detects the closed end (called “StallGuard”) is not working in this version of the firmware. During homing the motor will NOT stop by itself. You must watch the curtain and press Stop the moment it reaches the fully-closed position. If you do not stop it, the motor will keep running and strain against the stop. Always stay with the device while homing.

“Homing” teaches the device where the fully-closed position is. The motor drives the curtain toward the closed end; when it arrives, you stop it, and the device marks that spot as 0% (closed). Everything else — the open position, percentages, and the schedule — is measured from there.

Run homing when:

- You first install the device.
- The reported position no longer matches reality (for example after the rope slipped or was pulled by hand).

How to home (stop it yourself)

1. Start homing: tap Button 1 seven times quickly, or click the “Start-Stop Homing” button on the web page.

2. Watch the curtain drive toward the closed end.
3. The instant it reaches the fully-closed position, press Stop — press either button once, or click “Start-Stop Homing” again. This marks that spot as 0% (closed) and ends homing.

The “State” reading on the web page shows HOMING while homing is running and returns to IDLE once you press Stop.

Gentler alternative (no straining against the stop): use press-and-hold to inch the curtain to the fully-closed position, then hold both buttons together for 3 seconds. The device marks the current spot as home (0%) without driving into the end.

6. Set Up the Daily Schedule

The device can open and close the curtain automatically every day. There are two scheduling modes — pick the one you prefer:

- **Fixed times** — open and close at clock times you choose (e.g. 07:00 and 21:00).
- **Sunrise / sunset** — open at sunrise and close at sunset, so the curtain follows the daylight as it shifts through the seasons.

“Schedule Enabled” is the master on/off for whichever mode is active. The “Sun Schedule” switch chooses between the two: OFF uses the fixed times, ON uses sunrise/sunset. Both modes run on local time, so set your Timezone first (Section 7).

Fixed times (default)

Control	What to set
Open Time	The time of day to open the cover (e.g. 07:00).
Close Time	The time of day to close the cover (e.g. 21:00).
Sun Schedule	Leave this OFF to use fixed times.
Schedule Enabled	Turn this ON to run the schedule, OFF to pause it.
Timezone	Your local timezone, so the times mean what you expect (see Section 7).

Steps:

4. Set your Timezone first (Section 7) — the open/close times are in local time.
5. Set Open Time and Close Time.
6. Make sure Sun Schedule is OFF.
7. Turn Schedule Enabled ON.

Sunrise / sunset (Sun Schedule)

Turn “Sun Schedule” ON to follow the sun instead of the clock. The curtain opens at sunrise and closes at sunset for your location, recalculated every day so it tracks the changing seasons automatically — no need to adjust times by hand.

For this to be accurate, the device needs to know where it is. Enter your Latitude and Longitude in decimal degrees (north and east are positive; south and west are negative). You can copy these from any maps app — for example, New York is about Latitude 40.71, Longitude -74.01.

Control	What to set
Sun Schedule	Turn this ON to use sunrise/sunset.
Latitude	Your location's latitude in decimal degrees (e.g. 40.71).
Longitude	Your location's longitude in decimal degrees (e.g. -74.01).
Sun Open Offset	Minutes to shift the morning open relative to sunrise (see below).
Sun Close Offset	Minutes to shift the evening close relative to sunset (see below).
Schedule Enabled	Turn this ON to run the schedule.
Timezone	Your local timezone (see Section 7).

Offsets let you open or close a little before or after the sun, in minutes:

- A positive offset is after the event; a negative offset is before it.
- To open 30 minutes after sunrise, set Sun Open Offset to 30.
- To close 30 minutes before sunset, set Sun Close Offset to -30.
- Leave both at 0 to open exactly at sunrise and close exactly at sunset.

Steps:

8. Set your Timezone (Section 7).
9. Enter your Latitude and Longitude.
10. Optionally set the open/close offsets.
11. Turn Sun Schedule ON.
12. Turn Schedule Enabled ON.

Notes: Both schedules need an internet connection to keep accurate time. Scheduled moves are skipped while the device is homing, and every schedule setting is remembered across reboots. With Sun Schedule ON the fixed Open Time / Close Time are ignored. If you leave Latitude and Longitude at 0, sunrise/sunset will not match your location — set them before relying on the sun schedule.

7. Setting Your Timezone

The device keeps time using the internet, which provides time in UTC. To make the schedule fire at the correct local time, enter your timezone in the “Timezone” box using a POSIX timezone string. Copy the value for your region from the table in Section 8 and paste it into the Timezone box, then save.

Example: for U.S. Eastern time with daylight saving, enter EST5EDT,M3.2.0,M11.1.0.

Important about the number sign: In a POSIX timezone string the number is the hours WEST of UTC, so it looks “backwards” compared to the usual UTC offset. U.S. Eastern (UTC-5) is written

as 5, Pacific (UTC-8) as 8, and so on. The table in Section 8 already has this right — just copy it exactly, including capitalization and punctuation.

The trailing parts (for example “M3.2.0,M11.1.0”) are the daylight-saving rules. The U.S. switches on the 2nd Sunday of March and the 1st Sunday of November; the European Union switches on the last Sunday of March and the last Sunday of October. The tables in Section 8 already encode the correct rule for each region — just copy the value exactly. If your area does not observe daylight saving time, use the “without DST” value, which has no trailing rules.

8. Timezone POSIX Strings

Find your region below and copy the matching string into the Timezone box. Use the “With DST” column if your area changes its clocks twice a year; use “Without DST (standard all year)” if it does not.

United States

Zone	Where it's used	With DST	Without DST (standard all year)
Eastern	New York, Florida, Ohio, most of the East Coast	EST5EDT,M3.2.0,M11.1.0	EST5
Central	Texas, Illinois, Tennessee (western), most of the Midwest	CST6CDT,M3.2.0,M11.1.0	CST6
Mountain	Colorado, Utah, Montana, New Mexico	MST7MDT,M3.2.0,M11.1.0	MST7
Mountain (Arizona)	Most of Arizona — no daylight saving	(not used)	MST7
Pacific	California, Washington, Oregon, Nevada	PST8PDT,M3.2.0,M11.1.0	PST8
Alaska	Most of Alaska	AKST9AKDT,M3.2.0,M11.1.0	AKST9
Hawaii–Aleutian (Aleutian Is.)	Western Aleutian Islands, AK — observes DST	HST10HDT,M3.2.0,M11.1.0	HST10
Hawaii	State of Hawaii — no daylight saving	(not used)	HST10
Atlantic	Puerto Rico, U.S. Virgin Islands — no daylight saving	(not used)	AST4
Samoa	American Samoa — no daylight saving	(not used)	SST11
Chamorro	Guam, Northern Mariana Islands — no daylight saving	(not used)	ChST-10
Wake Island	Wake Island — no daylight saving	(not used)	<+12>-12

Zones marked “(not used)” under With DST do not observe daylight saving time, so only the standard-time string applies. Daylight-saving rules shown reflect current U.S. law (spring forward 2nd Sunday of March, fall back 1st Sunday of November).

Europe

Most of Europe observes EU daylight saving (forward on the last Sunday of March, back on the last Sunday of October). Countries that do not change their clocks are listed with a single standard-time value.

Zone	Where it's used	With DST	Without DST (standard all year)
Western European	Portugal (mainland), Canary Islands	WETWEST,M3.5.0/1,M10.5.0	WET0
UK / Ireland	United Kingdom, Ireland	GMT0BST,M3.5.0/1,M10.5.0	GMT0
Central European	France, Germany, Spain, Italy, Netherlands, Belgium, Poland, Sweden, Norway, Denmark, Switzerland, Austria, Czechia – most of central Europe	CET-1CEST,M3.5.0,M10.5.0/3	CET-1
Eastern European	Greece, Finland, Romania, Bulgaria, Estonia, Latvia, Lithuania, Ukraine	EET-2EEST,M3.5.0/3,M10.5.0/4	EET-2
Azores	Azores (Portugal)	<-01>1<+00>,M3.5.0/0,M10.5.0/1	<-01>1
Iceland	Iceland — no daylight saving	(not used)	GMT0
Kaliningrad	Kaliningrad, Russia – no daylight saving	(not used)	EET-2
Moscow	Moscow / western Russia — no daylight saving	(not used)	MSK-3
Belarus	Minsk, Belarus – no daylight saving	(not used)	<+03>-3
Turkey	Türkiye — no daylight saving	(not used)	<+03>-3

Zones marked “(not used)” under With DST do not change their clocks. POSIX values are taken directly from the IANA time zone database.

9. Resetting Wi-Fi

If you move the device to a new network, or want to hand it to someone else, you can erase the saved Wi-Fi details:

- Press and hold Button 3 for 3 to 10 seconds.
- The device erases its saved network and reboots.

- After rebooting it broadcasts the “ropener-XXXXXX” setup hotspot again — follow Section 2 to connect it to a network.

Note: This only clears Wi-Fi credentials. Your schedule, timezone, and tuning settings are kept.

10. Adjusting How It Moves (Advanced)

These settings are optional. The defaults work for a typical installation; change them only if needed. All values are remembered across reboots.

Setting	What it controls	Notes
Centimeters	Total travel distance of the curtain, in cm.	Set this to your curtain's open-to-closed travel. Default 30 cm.
Speed	How fast the motor runs (steps/second).	Higher is faster but louder. Effective max is 2000.
Acceleration	How quickly the motor speeds up / slows down.	Higher is snappier; too high can skip or be noisy.
Motor Direction	Which way the motor winds the rope.	Flip this if Open and Close are reversed for your install.
IRUN value	Motor running current / torque (1–31).	Raise for more pulling force; lower if the driver runs hot. Default 25.
SGTHRS value	Sensitivity of the stall detection used for homing (0–255).	See tuning note below.
TCOOLTHRS value	Speed threshold above which stall detection is active.	See tuning note below.

Tuning stall-based homing (optional)

Note: automatic stall detection is not working in the current firmware, so the SGTHRS / TCOOLTHRS settings have no effect yet — home manually by pressing Stop at the closed position (see Section 5). The guidance below applies once automatic detection is restored.

Homing is meant to detect when the motor meets resistance at the closed end. Two readings on the web page help you tune it:

- **SG_RESULT Sensor:** a live load reading that drops as the motor works harder.
- **TSTEP Sensor:** a measure of motor speed (smaller means faster).

To tune: run the cover under normal load while watching SG_RESULT, note the value it settles at, then set SGTHRS to roughly half of that. If homing stops too early (false stalls), lower SGTHRS; if it does not detect the stop, raise it. Leave these at their defaults unless homing is unreliable.

11. Home Assistant (Optional)

If you use Home Assistant, the device is discovered automatically over the local network and appears as a cover plus all the controls and readings in this guide. You do not need Home Assistant to use any feature — the web page offers the same controls.

When first added, Home Assistant may warn that communication is not encrypted. The device still works normally. To remove the warning, an encryption key can be enabled in the firmware configuration (a setup step for the installer).

12. Troubleshooting

Symptom	What to try
Can't find the web page	Make sure your phone/PC is on the same Wi-Fi as the device. Try the device's IP address instead of the .local name. Confirm it joined Wi-Fi (the setup hotspot should be gone).
No setup hotspot appears	Wait a full minute after power-on. If it already joined a network, hold Button 3 for 3-10 seconds to reset Wi-Fi and try again.
Open/Close are reversed	Toggle the "Motor Direction" switch.
Position is wrong / drifted	Run homing (Section 5) to recalibrate.
Schedule fires at the wrong time	Check the Timezone value (Section 7/8) and that the device has internet for time sync.
Schedule doesn't run	Confirm "Schedule Enabled" is ON and that Open/Close times are set.
Sun schedule opens/closes at the wrong time	Set Latitude and Longitude for your location and check the Timezone (Section 7/8). With Sun Schedule ON, the fixed Open/Close times are ignored.
Homing motor doesn't stop on its own	Expected in this version — automatic stop (StallGuard) is disabled. Press Stop when the curtain reaches the closed position (Section 5).
Motor driver gets hot	Lower the IRUN value.

13. Quick Reference

Control (web page)	Purpose
Ropener (cover)	Open, close, set position, or stop.
State	Shows IDLE / OPENING / CLOSING / HOMING.
Start-Stop Homing	Start or cancel calibration.
Open Time / Close Time	Daily schedule times (fixed-time mode).
Schedule Enabled	Turn the daily schedule on/off.
Sun Schedule	Switch between fixed-time and sunrise/sunset scheduling.
Latitude / Longitude	Your location used to compute sunrise/sunset.
Sun Open Offset / Sun Close Offset	Minutes to shift open/close relative to sunrise/sunset.
Timezone	Your local timezone (POSIX string).
Centimeters	Curtain travel distance.
Speed / Acceleration	Motion tuning.

Motor Direction	Reverse open/close direction.
IRUN value	Motor torque / current.
SGTHRS / TCOOLTHRS	Stall-detection tuning for homing.
SG_RESULT / TSTEP	Live readings used when tuning homing.
Physical button	Action
Button 1 – tap	Close (or stop if moving).
Button 1 – hold	Close while held (stops on release and at the closed end).
Button 1 – seven quick taps	Start homing.
Button 2 – tap	Open (or stop if moving).
Button 2 – hold	Open while held (stops on release and at the open end).
Both buttons – hold 3 sec	Set current position as home (0%).
Button 3 – long press	Reset Wi-Fi and reboot to setup.